Technical review

Item #2 · 12-mailbox.mbox · Run ID: 20261113T232800Z-08b8d1

Path	`R:\evidence\mail\12-mailbox.mbox`
Size (bytes)	3238
Last modified (UTC)	2026-11-12T06:00:00Z
Extension	.mbox
Detected type (signature)	Mailbox (mbox)
Status	No issues found
Case	RV-SAMPLE-001
Evidence ID	E0016
Case registry	Verified: identical to the registered evidence
Registered at intake (UTC)	2026-11-13T23:27:52.795Z

Hashes

sha256	`2e39de637dad0f16469632facf7544dfe4fc0d39bf7210dd579b7a74ba0f5ebe`
md5	`f4e8e05ccc50d543ed3703f21cb965d4`

Findings

None

Email analysis

Messages in this mbox	3
#1 @0	R:\evidence\mail\12-mailbox.mbox#1@0 (1321 B) · From bruno@example.org Wed Nov 11 11:00:03 2026
#2 @1372	R:\evidence\mail\12-mailbox.mbox#2@1372 (1276 B) · From bruno@example.org Wed Nov 11 11:30:02 2026
#3 @2699	R:\evidence\mail\12-mailbox.mbox#3@2699 (490 B) · From bruno@example.org Wed Nov 11 09:40:00 2026

Method

Each file was read once to compute its digests, then analysed as email: .eml as an RFC 5322 message, .msg through its MAPI properties, and each message of an mbox as an item of its own (each in a separate worker process with time and memory limits when worker isolation is on, as recorded in the audit log). The header section, Received hops and dates were read and compared; DKIM and ARC signatures were verified offline with the public keys the examiner supplied (no DNS query unless online mode was chosen); Authentication-Results were recorded as stated evidence; attachments were hashed; and BEC heuristics compared the addresses with the organisation's own domains. Every step and every file written is recorded in the hash-chained audit log.

Audit trail

Audit entries	#13 – #18
Log folder	`R:\case\runs\rv-mailauth\20261113T232800Z-08b8d1\logs`
Digital signature	This PDF is not digitally signed.

Limitations

- DKIM results depend on the public keys supplied: a check is only meaningful with the key that was valid when the message was sent. Keys are rotated, so a current key can fail a genuine message and a missing key leaves the signature unchecked.
- A missing DKIM signature is common and proves nothing; a valid signature covers only the signed header fields and the body, not who wrote the message.
- Outlook .msg files do not keep the original message: the body is re-encoded (so DKIM cannot be checked) and the original header section is kept only when transport headers were stored.
- Lookalike-domain, Reply-To, display-name and payment-wording checks are heuristics with false positives and false negatives; the organisational domain comes from the bundled Public Suffix List, whose version is recorded in environment.json and may differ from the list in force when the message was sent.
- Received, Date and Authentication-Results headers are text that servers and clients wrote; unsigned headers can be added or edited by anyone who handled the file.
- In an mbox, body lines starting with "From " are escaped with '>'; un-escaping follows the selected variant (mboxrd by default) and cannot always restore the original line, which can affect DKIM body hashes.
- A result without findings does not prove that a file is authentic or unaltered before acquisition.
- Modification times come from the file system of the examined copy and may differ from the original source.
- Type detection uses a limited table of common file signatures; unknown types are reported as unknown, not as suspicious.
- A trusted timestamp proves the audit log existed in this exact form at the stated time; it does not prove that the examined copy matches the original source.